PERMANENCE OF LOCAL FINITE PRESENTATION AND LOCAL FINITE TYPE FOR THE THIRD SIDE OF A TRIANGLE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A flat morphism locally of finite presentation is open. Morphisms, Lemma morphisms-lemma-fppf-open, invoked at proof lines 4623-4624. Lets finitely many affine opens of X cover the quasi-compact affine Y.
- The algebra statement for permanence of flat finite presentation (and of finite type) along a surjective flat finitely presented map of affines. Same chapter, Lemmas lemma-flat-finitely-presented-permanence-algebra and lemma-finite-type-local-source-fppf-algebra, invoked at proof lines 4629-4632. Gives the result in the affine case, to which the reduction has brought us.

Source reference. Original lemma 0367: sources/stacks-descent/pinned-original-source.tex, source lines 4569–4569. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KK: sources/stacks-descent/pinned-original-source.tex, source lines 4478–4478. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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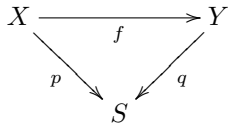
Source reference. Original lemma 01UA: sources/stacks-chow/tags.txt, source lines 2403–2403. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let*



be a commutative diagram of morphisms of schemes. Assume that f is surjective, flat and locally of finite presentation and assume that p is locally of finite presentation (resp. locally of finite type). Then q is locally of finite presentation (resp. locally of finite type).

Proof. The problem is local on S and Y. Hence we may assume that S and Y are affine. Since f is flat and locally of finite presentation, we see that f is open (Morphisms, Lemma 01UA). Hence, since Y is quasi-compact, there exist finitely many affine opens $X_i \subset X$ such that $Y = \bigcup f(X_i)$. Clearly we may replace X by $\coprod X_i$, and hence we may assume X is affine as well. In this case the lemma is equivalent to Lemma 02KK (resp. Lemma 0367) above. □

References

[EGA] Dieudonné, Jean and Grothendieck, Alexander. Éléments de géométrie algébrique IV. Inst. Hautes Études Sci. Publ. Math., volumes 20, 24, 28, 32, 1964-1967.